Reg. No. | | | | | | | | | |

Question Paper Code : 15717

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2019

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. – Information Technology

15MA304 – MATHEMATICS III – DISCRETE MATHEMATICS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

Cog.		
U, CO1	1.	If A is a finite set with 3 elements, how many relations are there on A?
U,CO1	2.	Of 43 students in a village who applied for engineering or medical or for both, 32 applied for engineering and 19 applied for medical. Find the number of students who applied only for medical.
A,CO2	3.	Consider the statement $\neg p \Rightarrow \neg q$ *is* logically equivalent to which of the statement given below? Justify your answer. $(i)\ p \Rightarrow q$ $(ii)\ q \Rightarrow p$ $(iii)\ \neg q \vee p$ $(iv)\ \neg p \vee q$ A) i only B) i and iv only C) ii only D) ii and iii only
U,CO2	4.	Let $p(x)$ denote the statement "$x > 3$". The truth value of $p(4)$ is _____. Justify.
U,CO3	5.	Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE? Justify. P: Minimum spanning tree of G does not change Q: Shortest path between any pair of vertices does not change A) P only B) Q only C) Neither P nor Q D) Both P and Q
U,CO3	6.	State Hand-Shaking theorem.

A,CO4 7. If the homomorphism $f:(G,*)\to(G',\Delta)$ is onto, then f is called _____. Justify.

A) Epimorphism B) Monomorphism

C) Isomorphism D) Endomorphism

R,CO4 8. Give an example of a ring which is not a field.

U,CO5 9. Define a distributive lattice.

U,CO5 10. The dual of $(a+a).(b+1)=a$ is _____. Justify.

PART B – (5 × 16 = 80 Marks)

A,CO1 11. a) i. If A= {1,2,3}, B= {3,4,5}, C={4,5,6} then find A) A×(B∩C)

B) (A×B)∩(A×C) C) A× (BUC) D) (A×B)U(A×C) (8 Marks)

L,CO1 11. a) ii. Let A= {1,2,3,4,5,6,7} and relation R defined as

R= $\{(x,y)/\ |x-y|=2\}$. Verify R is an equivalence relation. (8 Marks)

OR

A,CO1 11. b) i. R and S are "Congruent modulo 3" and "Congruent modulo 4" relations respectively on the set of integers. That is,

$R=\{(a,b)/\ a\equiv b(\text{mod}\,3)\}$

$S=\{(a,b)/\ a\equiv b(\text{mod}\,4)\}$

Find RUS, R-S, S-R, R∩S. (8 Marks)

L,CO1 11. b) ii. If f:R→R and g: R→R are defined as $f(x)=x^2-2$ *and* $g(x)=x+4$. Find $f\bullet g,\ g\bullet f$ and state whether these functions are injective, surjective and bijective. (8 Marks)

A,CO2 12. a) i. Show that the following premises are inconsistent.

A) If Nirmala misses many classes through illness, then she fails high school.

B) If Nirmala fails high school, then she is uneducated.

C) If Nirmala reads a lot of books, then she is not uneducated.

D) Nirmala misses many classes through illness and reads a lot of books. (8 Marks)

A,CO2 12. a) ii. Prove that the following premises $\exists x(p(x)\wedge Q(x)),\forall x(Q(x)\to R(x))$ imply the conclusion $\exists x(p(x)\wedge R(x))$. (8 Marks)

OR

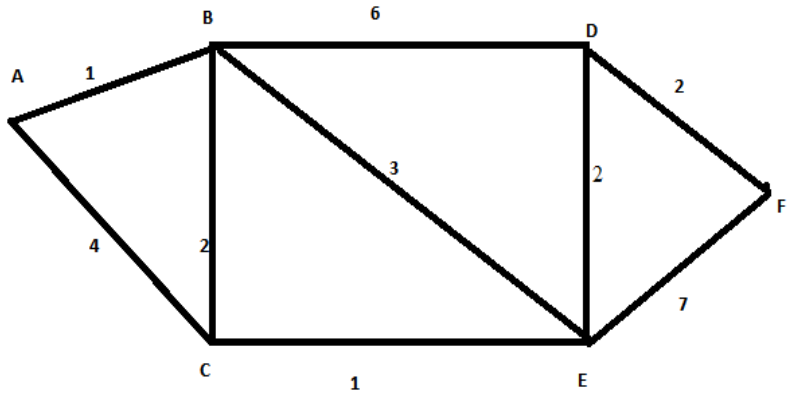
A,CO2 12. b) i. Show that the premises

"All integers are rational numbers". "Some integers are powers of 2". Imply the conclusion "Some rational numbers are powers of 2". (8 Marks)

A,CO2 12. b) ii. Using rules of inference, show that $s \vee r$ is tautologically implied by $p \vee q, p \to r, q \to s.$ (8 Marks)

A,CO3 13. a) i. Prove that the maximum number of edges in a simple disconnected graph G with n vertices and k components is $\frac{(n-k)(n-k+1)}{2}$. (8 Marks)

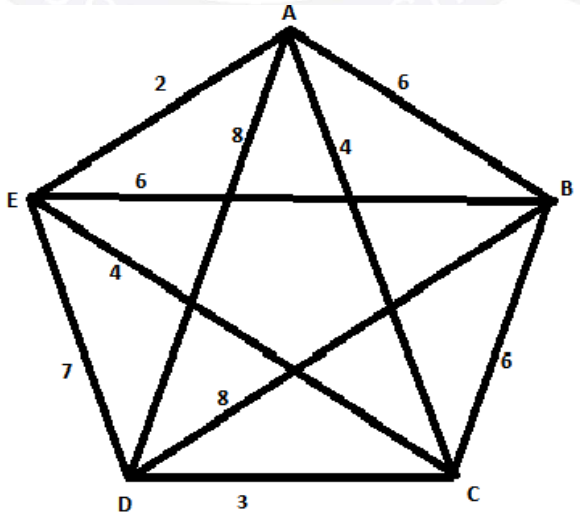
A,CO3 13. a) ii. Use Dijkstra's algorithm to find the shortest path between A and F in the given weighted graph.



(8 Marks)

OR

A,CO3 13. b) i. Find the minimum spanning tree for the given weighted graph using Kruskal's algorithm.



(8 Marks)

A,CO3 13. b) ii. Give an example with explanation of a graph which contains

A) an Eulerian circuit that is also a Hamiltonian circuit

B) an Eulerian circuit, but not a Hamiltonian circuit

C) an Hamiltonian circuit, but not an Eulerian circuit (8 Marks)

A,CO4 14. a) i. Let G be the set of all matrices of the form $\begin{pmatrix} x & x \\ x & x \end{pmatrix}, x \in R^*$ where R^* is a set of real numbers except 0. Show that G is a group under matrix multiplication. (8 Marks)

U,CO4 14. a) ii. Prove that the set of all semi group automorphisms of a semi group is a semi group under the operation of left composition. (8 Marks)

OR

A,CO4 14. b) i. Prove that the necessary and sufficient condition for a non-empty subset H of a group {G,*} to be a subgroup is $a, b \in H$ which implies $a * b^{-1} \in H$. (8 Marks)

U,CO4 14. b) ii. Prove that every subgroup of a cyclic group is cyclic. (8 Marks)

A,CO5 15. a) i. Show that every chain is a distributive lattice. (8 Marks)

A,CO5 15. a) ii. Let (L, ≤) be a lattice and $a, b \in L.$ Show that

$$a \le b \Leftrightarrow a \vee b = b \Leftrightarrow a \wedge b = a$$ (8 Marks)

OR

A,CO5 15. b) i. In a Boolean lattice, state and prove the De Morgan's Laws. (8 Marks)

A,CO5 15. b) ii. Prove that $a \wedge b \le a,\ b \le a \vee b$ for any two elements a and b in a lattice (L, ≤). (8 Marks)